

Single Core Copper Conductor, XLPE insulated

Copper Wire Screen and HDPE Sheath
(Cu/XLPE/CW/HDPE)

Voltage Grade

26 / 45 (52) KV

Description

Stranded circular or segmental compacted copper conductor Filled with swelling powder between conductor layers, Semi-conductive water blocking tape over conductor to protect the conductor from longitudinal water penetration, semi-conducting layer as conductor screen, XLPE insulated, semi-conducting layer as non metallic insulation screen, Semi-conductive water blocking tape to protect the screen area from longitudinal water penetration, copper wire as metallic insulation screen to withstand the required earth fault current, water blocking tape to protect the screen area from longitudinal water penetration, copolymer aluminum tape to protect the cable from radial water penetration and HDPE sheathed with graphite coating or extruded semi-conducting layer for the D.C Sheath test.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

Relevant Standard	IEC 60228, 60840, IEC 60229 and 60811
Conductor	Plain copper conductor class 2 Circular Comnpacted (applied for C.S.A up to 800 mm2) or Segmental Compacted (Applied from C.S.A = 1000 mm2 and above) Five segments of compacted conductor in sector shape of 72° are assembled together with separation of non-metallic tapes to reduce the skin effect which reduce the Rac.
Conductor Screen	Extruded Semiconductive Compound
Insulation	Extruded XLPE Compound Rated 90° C
Insulation Screen	Non Metallic part: Extruded Semiconductive Compound (Bonded) to the insulation Metallic part: Copper Wires helically applied and banded with a Copper tape to achieve electrical contact Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.
Outer Sheath	An extruded layer of HDPE shall be applied over the metallic layers for protection against chemical reaction & mechanical protection. Extruded semi conductive layer or graphite coating shall be applied over the outer sheath for D.C testing purpose
Water Pentration layers	i) Longitudinal Water Penetration: 1) the conductor: the conductor layers filled with swelling powder and overlaped semiconductive water blocking tape over the conductor 2) The Metallic part of the insulation screen: An overlaped semiconductive water blocking tape before the metallic screen & an overlaped water blocking tape after the metallic screen ii) Radial Water Penetration: Copolymer aluminum tape applied Longitudinally before the PE outer Sheath
Outer Sheath Color	Black (default color), or any other color according to customer request

Note:-

Flame retardant PVC sheathed or MDPE, LLDPE Sheathed are available according to customer request
The Metallic Screen is designed to carry an earth fault cureent = 25 KA per one seond per three pahses as default, any other design is also available as per customer request



HIGH VOLTAGE CABLES

Maximum Operating Temp. 90° C

Minimum bending Radius (16-18) x outer diameter

Packing Condition Our standard length on Steel Drum / or according to customer request

Product Code	Nominal Cross Sectional Area	Conductor			Operating Capacitance	Inductance		Thickness of Conductor Screen	Thickness of Insulation	Thickness of Conductor Screen	Thickness of Insulation	Approx. Overall Diameter	Approx. Weight
		DC at 20 °C	AC at 90 °C			Flat seperated	Trefoil						
			Flat seperated	Trefoil									
	mm²	Ω/km	Ω/km	Ω/km	μF/km	mH/km	mH/km	mm	mm	mm	mm	mm	kg/km
Electrical & Constrctional data (Nominal Values)													
PH-CG15-XA-01-BH	70 R	0.2680	0.3419	0.3420	0.1516	0.6657	0.4808	0.6	9.0	0.8	2.1	40.9	2180
PH-CG16-XA-01-BH	95 R	0.1930	0.2464	0.2465	0.1639	0.6443	0.4594	0.6	9.0	0.8	2.2	42.6	2485
PH-CG17-XA-01-BH	120 R	0.1530	0.1955	0.1957	0.1753	0.6274	0.4425	0.6	9.0	0.8	2.2	44.0	2755
PH-CG18-XA-01-BH	150 R	0.1240	0.1587	0.1588	0.1859	0.6146	0.4298	0.6	9.0	0.8	2.3	45.5	3075
PH-CG19-XA-01-BH	185 R	0.0991	0.1271	0.1274	0.2012	0.5947	0.4099	0.6	9.0	0.8	2.3	47.4	3485
PH-CG20-XA-01-BH	240 R	0.0754	0.0970	0.0974	0.2204	0.5773	0.3925	0.6	9.0	0.8	2.4	50.0	4110
PH-CG21-XA-01-BH	300 R	0.0601	0.0778	0.0783	0.2379	0.5640	0.3791	0.6	9.0	0.8	2.5	52.4	4770
PH-CG22-XA-01-BH	400 R	0.0470	0.0614	0.0622	0.2593	0.5486	0.3638	0.6	9.0	0.8	2.6	55.3	5675
PH-CG23-XA-01-BH	500 R	0.0366	0.0486	0.0498	0.2853	0.5343	0.3494	0.6	9.0	0.8	2.7	58.8	6830
PH-CG24-XA-01-BH	630 R	0.0283	0.0386	0.0401	0.3136	0.5213	0.3365	0.6	9.0	0.8	2.8	62.6	8215
PH-CG25-XA-01-BH	800 R	0.0221	0.0313	0.0332	0.3511	0.5108	0.3260	0.8	9.0	0.8	2.9	67.6	10165
PH-CM26-XA-01-BH	1000 S	0.0176	0.0247	0.0252	0.4128	0.4991	0.3142	1	9.0	1	3.1	76.3	12910
PH-CM27-XA-01-BH	1200 S	0.0151	0.0219	0.0224	0.4330	0.4940	0.3092	1	9.0	1	3.2	79.1	14330

Product Code	Nominal Cross Sectional Area	Laying conditions: Trefoil formation			Laying conditions: Flat formation		
		Type of Earthing Bonding System	Direct burial	in Air (Shaded)	Type of Earthing Bonding System	Direct burial	in Air (Shaded)
							
			$p_t=120$ $T=35\text{ }^{\circ}\text{C}$	$T=40\text{ }^{\circ}\text{C}$		$p_t=120$ $T=35\text{ }^{\circ}\text{C}$	$T=40\text{ }^{\circ}\text{C}$
Continuous current ratings (load factor = 100%) for one circuit in operation (Amperes)							
PH-CG15-XA-01-BH	70 R	Double End Bonded	223	279	Single End Bonded or Cross Bonding	235	321
PH-CG16-XA-01-BH	95 R		265	337		281	390
PH-CG17-XA-01-BH	120 R		299	387		319	449
PH-CG18-XA-01-BH	150 R		334	436		358	509
PH-CG19-XA-01-BH	185 R		376	499		405	586
PH-CG20-XA-01-BH	240 R		432	582		471	693
PH-CG21-XA-01-BH	300 R		484	666		532	796
PH-CG22-XA-01-BH	400 R		544	765		607	924
PH-CG23-XA-01-BH	500 R		610	877		692	1076
PH-CG24-XA-01-BH	630 R		680	1000		788	1251
PH-CG25-XA-01-BH	800 R		735	1130		889	1442
PH-CM26-XA-01-BH	1000 S		858	1338		1024	1700
PH-CM27-XA-01-BH	1200 S		906	1431		1098	1842

The above data is approximate and subject to manufacturing tolerances